

Sets

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

Distributive law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Verify the distributive law for:

$$A = \{1, 2, 3\}, B = \{2, 4, 6\}, C = \{4, 8\}$$

First part

Left hand side: $A \cup (B \cap C) =$

Right hand side: $(A \cup B) \cap (A \cup C) =$

Second part

Left hand side: $A \cap (B \cup C) =$

Right hand side: $(A \cap B) \cup (A \cap C) =$